

Higher Diploma in Science in Computer Science

Overview of Structure & Purpose of the Online Materials

Tutors



Slack



Moodle



Youtube



4 Key Web
Services

Tutors



Github ID

Slack



personal email

Moodle



wit email

Youtube



open

Programme homepage

<https://tutors.dev/course/setu-hdip-comp-sci-2025>

**Higher Diploma in Computer Science 2025**
Department of Computing & Mathematics, SETU





Semester One - January - June 2025

FAQ
Questions and Answers for new students on the programme

Workshop One
induction · structure ·
schedules · handbook

Life etc...
time · space · balance ·
experiences · tips · techniques

Web Development I
html · css · layout · web apps ·
web frameworks · templating

Programming
algorithms · data structures ·
processing · java · classes ·
libraries

Links to individual module
Home pages

A course homepage

Full Stack Web Development
Eamonn de Leastar

Search

Layout

0

Full Stack Web ...

Overview +
Assignments



Overview, schedule & assignments.

01: Node



The fundmendals of node.
A first node application.

02: Hapi.js



The Hapi application
Framework

03: Joi



Validation with Joi

04: TDD



Test Driven Development

05: Models



Incorporate Mongo Models

06: APIs



Expose an API from an
application

07: REST



Resful API implementations

08: OpenAPI



Documentating
OpenAPI/Swagger APIs

09 JWT



Authenticating APIs with
JSON Web Tokens

10: Seeding



Seeding the Database

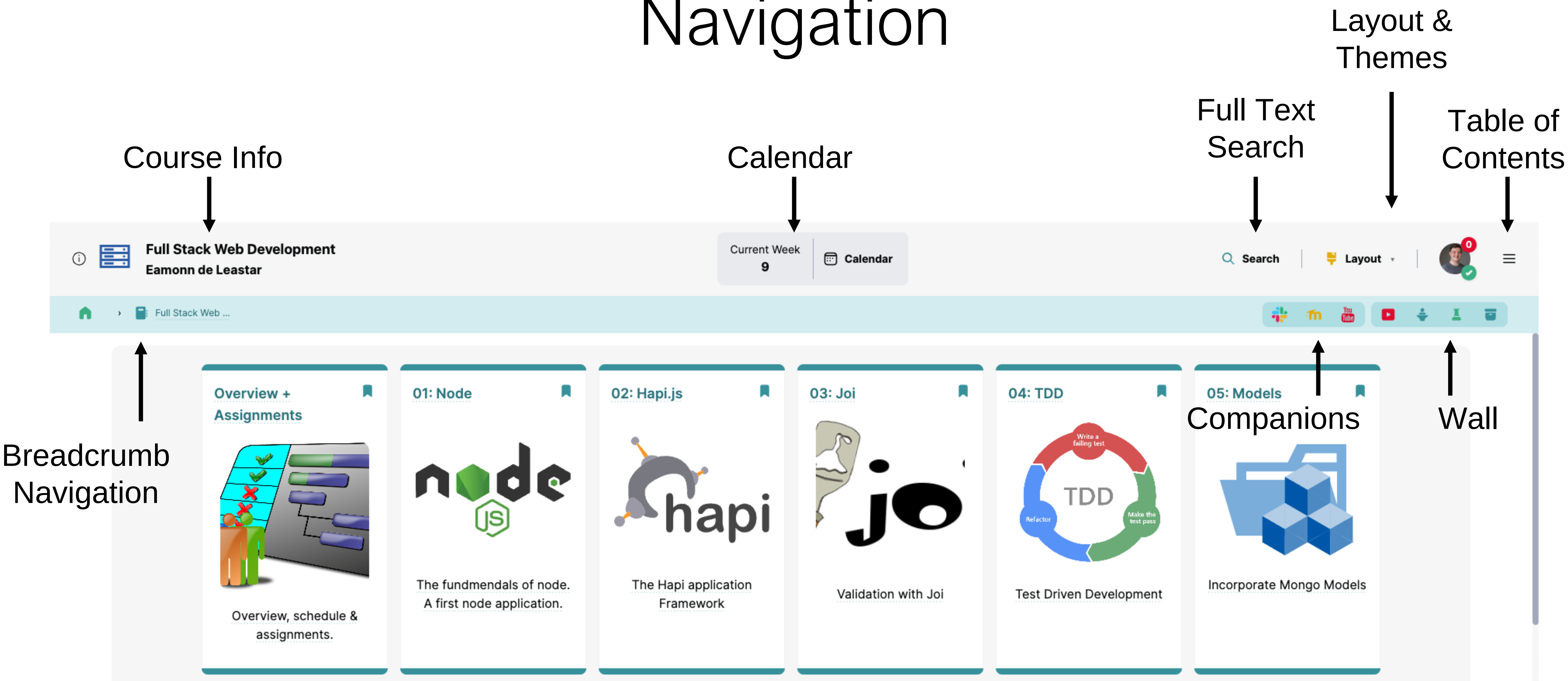
11: Deployment &
Images



Deploying to Monogo
Cloud, Heroku &
Cloudinary

Topics - 1 topic/card for each week

Navigation



Navigation

Layout & Themes



The screenshot displays a web application interface for 'Full Stack Web Development' by Eamonn de Leastar. The top navigation bar includes a search bar, a 'Layout' toggle, and a user profile. Below the navigation bar is a grid of course modules. A 'Layout & Themes' dropdown menu is open on the right side of the grid.

Navigation Bar:

- Full Stack Web Development
Eamonn de Leastar
- Current Week 9
- Calendar
- Search
- Layout
- User Profile

Course Modules:

- Overview + Assignments: Overview, schedule & assignments.
- 01: Node: The fundmendlals of node. A first node application.
- 02: Hapi.js: The Hapi application Framework
- 03: Joi: Validation with Joi
- 04: TDD: Test Driven Development
- 05: Incorporate Mongo Models

Layout & Themes Dropdown:

- Toggles
 - Dark Mode
 - Compact
- Themes
 - Tutors
 - Dyslexia

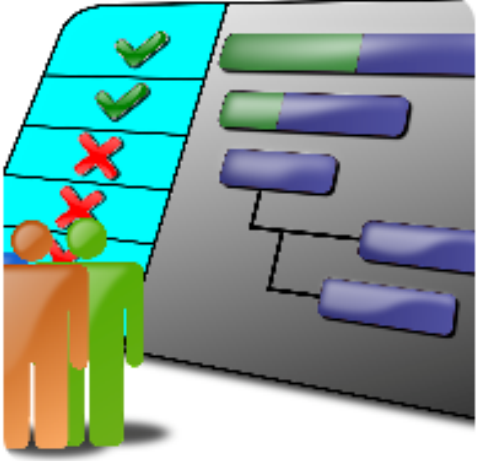
Theme Support

Layout &
Themes



The screenshot displays a web application interface in dark mode. The top navigation bar includes a user profile section for 'Full Stack Web Development' by 'Eamonn de Leastar', a search bar, and a 'Layout' dropdown menu. The 'Layout' menu is open, showing options for 'Toggles' (Dark Mode, Compact, Toggle Light Mode) and 'Themes' (Tutors, Dyslexia). The main content area features a series of six cards: 'Overview + Assignments', '01: Node', '02: Hapi.js', '03: Joi', '04: TDD', and '05: Incorporate Mongo Models'. Each card contains a title, a description, and a visual element (e.g., Node.js logo, Hapi.js logo, Joi logo, TDD diagram). The interface is designed to be accessible, with a focus on theme and layout customization.

Topics - 1 topic/card for each week

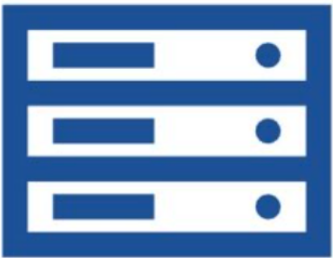
Overview + Assignments  Overview, schedule & assignments.	01: Node  The fundmendlals of node. A first node application.	02: Hapi.js  The Hapi application Framework	03: Joi  Validation with Joi	04: TDD  Test Driven Development	05: Models  Incorporate Mongo Models
06: APIs  Expose an API from an application	07: REST  Resful API implementations	08: OpenAPI  Documentating OpenAPI/Swagger APIs	09 JWT  Authenticating APIs with JSON Web Tokens	10: Seeding  Seeding the Database	11: Deployment & Images  Deploying to Monogo Cloud, Heroku & Cloudinary

Single
Topic

15

Watch Later

Share



Higher Diploma in Computer Science

Full Stack Web Development



Watch on

YouTube

Topic Video

Pre-recorded or live stream

Routing & Authentication

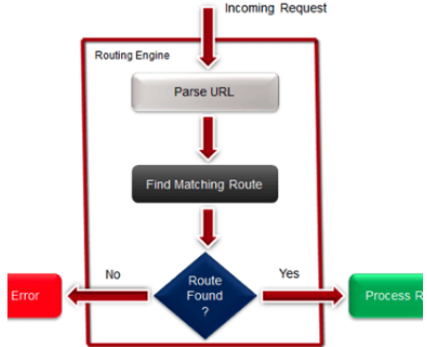
Routing



Routing



HOW ROUTING WORKS



Hash based routing in Svelte

Pages & Components





Creating and assembling Svelte Components

Lab-15a-Donation-Svelte-1





Front end for Donation - Pages + Components

Sub Topic 1

Sub Topic

Svelte Fundamentals



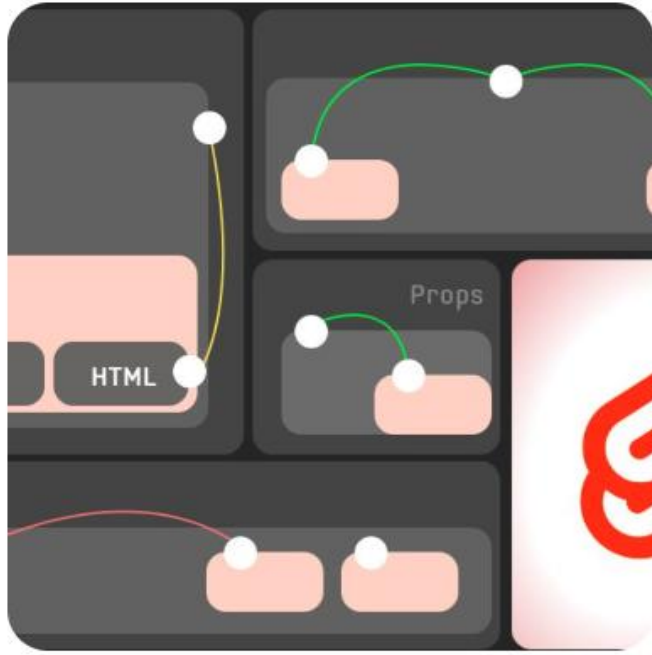
Single
Video

Svelte Core Concepts



Essential features of the Svelte Framework

Svelte Components



Creating and assembling Svelte Components

Lab 14a-Todo 2



Explore Svelte components in the Todo app.

Svelte Handbook

THE SVELTE HANDBOOK



The book is available at <https://flaviocopes.com/pa>

Talk Slide PDFs

Lab

Embedded
PDF reader
custom
with
Navigation

4 of 18

Context

The context API is provided by 2 functions which are provided by the svelte package: `getContext` and `setContext`

You set an object in the context, associating it to a key

```
<script>
import { setContext } from 'svelte'

const someObject = {}

setContext('someKey', someObject)
</script>
```

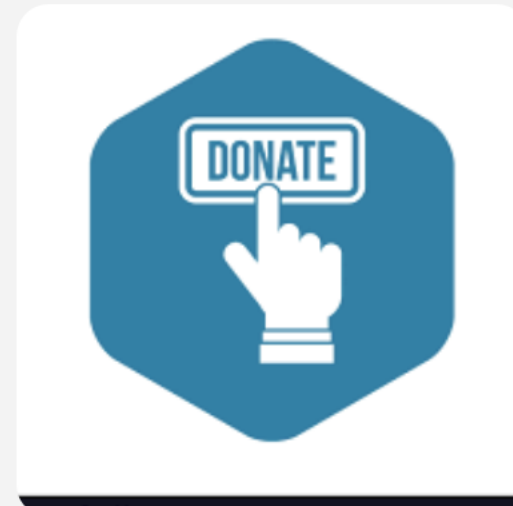
```
<script>
import { getContext } from 'svelte'

const someObject = getContext('someKey')
</script>
```

4

Talk

16: Svelte Applications



 Svelte Applications

 Svelte Applications

 Why Svelte? 

 Donate API 

 Donate in Svelte 

 Lab-16a-Donate-Hapi-3 

 Lab-16b-Donate-Svelte-3 

Talk
Context -
related
talks &
videos

Labs

Lab Steps



Objectives

Create

Router

Components

Pages

Donate & Report

Login & Signup

Solutions

Router

Like web apps, front end applications also have the concept of a router. Svelte itself does not provide one, but there are several popular options:

- <https://www.routify.dev/>
- <https://github.com/AlexxNB/tinro>
- <https://github.com/ItalyPaleAle/svelte-spa-router>

We will use the last one - which is documented in detail in this svelte text book:

- [Svelte 3 Up & Running](#)

Start by installing it:

```
npm install svelte-spa-router
```

To try it out, create three svelte file in a new **pages** folder in the project:

pages/Main.svelte

```
<div class="notification is-link">
  Hello from Main!
</div>
```

pages/Login.svelte

```
<div class="notification is-info">
  Hello from Login!
</div>
```

pages/Signup.svelte

```
<div class="notification is-warning">
  Hello from signup!
</div>
```

This is a revised App.svelte, to include the routing table:

App.svelte

Markdown
formatted text



Code block
syntax
highlighting



Search

Enter search term:

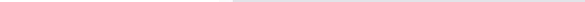
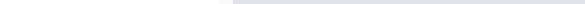
env

<pre>"env": {</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools	<pre># dotenv environment variables file</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools	<pre># dotenv environment variables file</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools
<pre>.env</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools	<pre>.env.test</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools	<pre>.env.production</pre> Lab-01-Playtime-0-1-0/ Code Quality Tools
In general we should attempt to "externalise" sensitive information, secrets or API details to a text file. This is usually called <code>.env</code> Lab-02-Playtime-0-2-0/ Exercises	<pre>.env</pre> Lab-02-Playtime-0-2-0/ Exercises	<pre>https://github.com/motdotla/dotenv#readme</pre> Lab-02-Playtime-0-2-0/ Exercises
In general we should attempt to "externalise" sensitive information, secrets or API details to a text file. This is usually called <code>.env</code> Lab-03-Playtime-0-3-0/ Exercise 3	<pre>.env</pre> Lab-03-Playtime-0-3-0/ Exercise 3	<pre>https://github.com/motdotla/dotenv#readme</pre> Lab-03-Playtime-0-3-0/ Exercise 3
<pre>npm install dotenv</pre> Lab-03-Playtime-0-3-0/ Exercise 3	Import the package in server, and check for the presence of the <code>.env</code> file, exiting if not found: Lab-03-Playtime-0-3-0/ Exercise 3	<pre>import dotenv from "dotenv";</pre> Lab-03-Playtime-0-3-0/ Exercise 3
<pre>import dotenv from "dotenv";</pre> Lab-03-Playtime-0-3-0/ Exercise 3	<pre>const result = dotenv.config();</pre> Lab-03-Playtime-0-3-0/ Exercise 3	If we have included the <code>.env</code> file above at the root of our project, then we can directly access its entries. So, our cookie configuration can now import these values:

Youtube Live



Live Webinars
Delivery

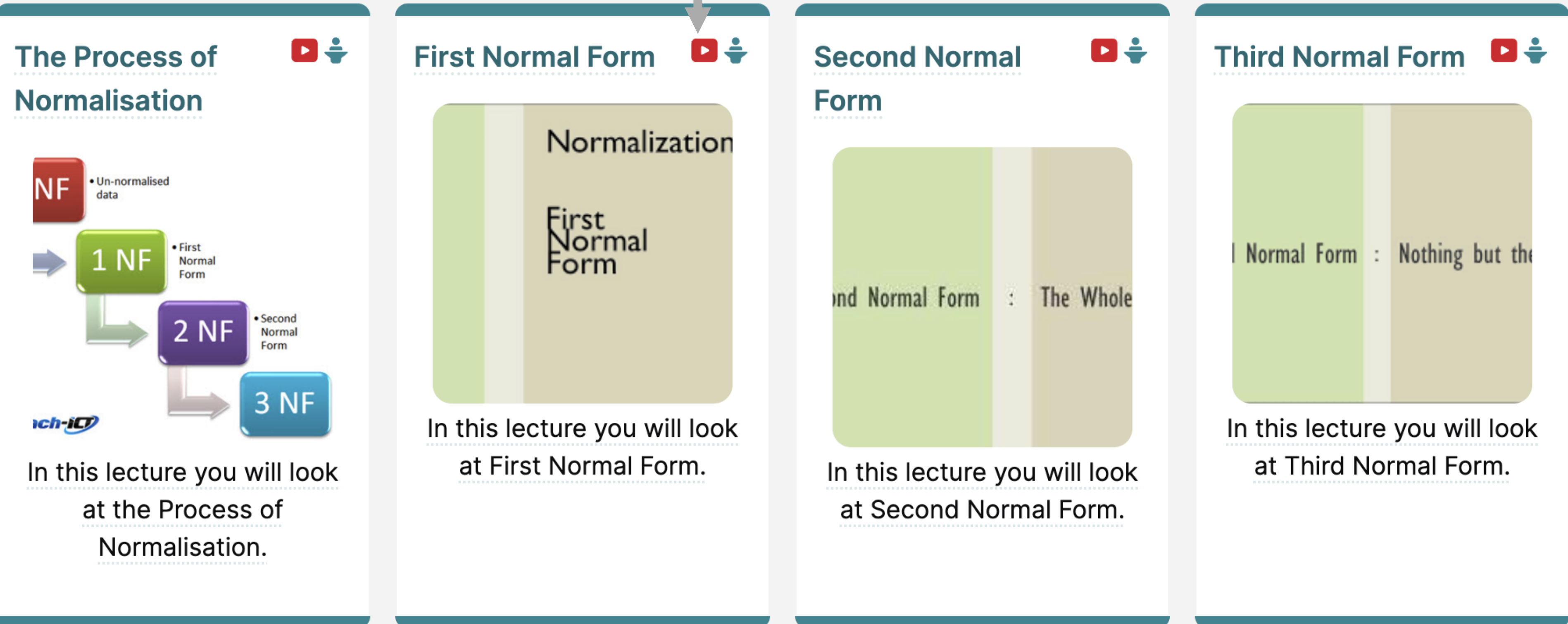


The image is a vertical composition of three distinct parts. The top part shows a web browser header for "Routing Full Stack Web Development". It includes a search bar, a "Layout" dropdown menu, and a user profile icon. Below the header is a breadcrumb trail: "Full Stack Web ..." > "15: Routing & A..." > "Routing". To the right of the breadcrumbs are several social media icons (Discord, GitHub, YouTube) and utility icons (play, share, etc.). The middle part is a screenshot of a YouTube video titled "Device/Service Discovery" from a channel named "2021 Computer Systems and Networks". The video features a blue title banner, a network diagram, and a bulleted list of protocols: DHCP, SDP, UDDI, and SSDP. A small inset shows a man speaking. The video player controls at the bottom indicate it's at 4:33 / 1:36:38. The bottom part contains two side-by-side cards. The left card, titled "MQTT and HTTP", compares MQTT (Publish-Subscribe) with HTTP (Web APIs). The right card, titled "Lab Messaging", illustrates using MQTT to connect devices via a central broker.



Pre-recorded Videos for selected slides & labs

Stages of Normalisation



Moodle



WIT E-Learning
Platform

Used for Assignment
Submission +
feedback + grades

Occasionally used for
additional document
distribution



 **moodle.wit**

 **Higher Diploma in
Computer Science**

 Participants

 Grades

Grades

Higher Diploma in Computer Science

[My Home](#) / [My modules](#) / [Higher Diploma in Computer Science](#)

Programming



-  [Assignment 1](#)
-  [Assignment 2](#)
-  [Assignment 3](#)

Web Development



-  [Assignment 1](#)
-  [Assignment 2](#)

Skills Studio 1



-  [Assignment](#)

Assignment
Submissions



Assessment feedback

Web Development



 Assignment 1

 Assignment 2

Assignment: Submit

https://moodle.wit.ie/mod/assign/view.php?id=2407050&rownum=0&action=grader&userid=42131

Module: Higher Diploma in Computer Science

Assignment: Submit

View all submissions



@mail.wit.ie

Change user

Due date: 11 March 2018, 11:55 PM

33 of 39

Well done his for a excellent effort - your grade was 74%

The structure of the project was excellent, with effective use of file and folder names. Indentation was good (although head and body were not indented) and the use fo semantic elements was spot on. You implemented 2 column layout, which worked well. Navigation was also excellent - with full context sensitivity. The CSS was clean and concise. I was rewarding an attempt to use more than one sheet - a generic and a set of styles (perhaps that could have been used on different sites).

You didn't attempt EJS. Templating is important in the remainder of the course (and in successor modules) - so it would be worth, perhaps, having a go a the template labs if you have time.

I didn't see any media, but I was able to verify a deployment

Overall I was pleased with the projects - in my categorisation there were 7 outstanding projects, 10 excellent, 6 very good, 4 good and 1 fair. The were graded more or less adhering to the published grading spectrum.

Notify students ☐

Save changes

Save and show next

Reset

Tutors



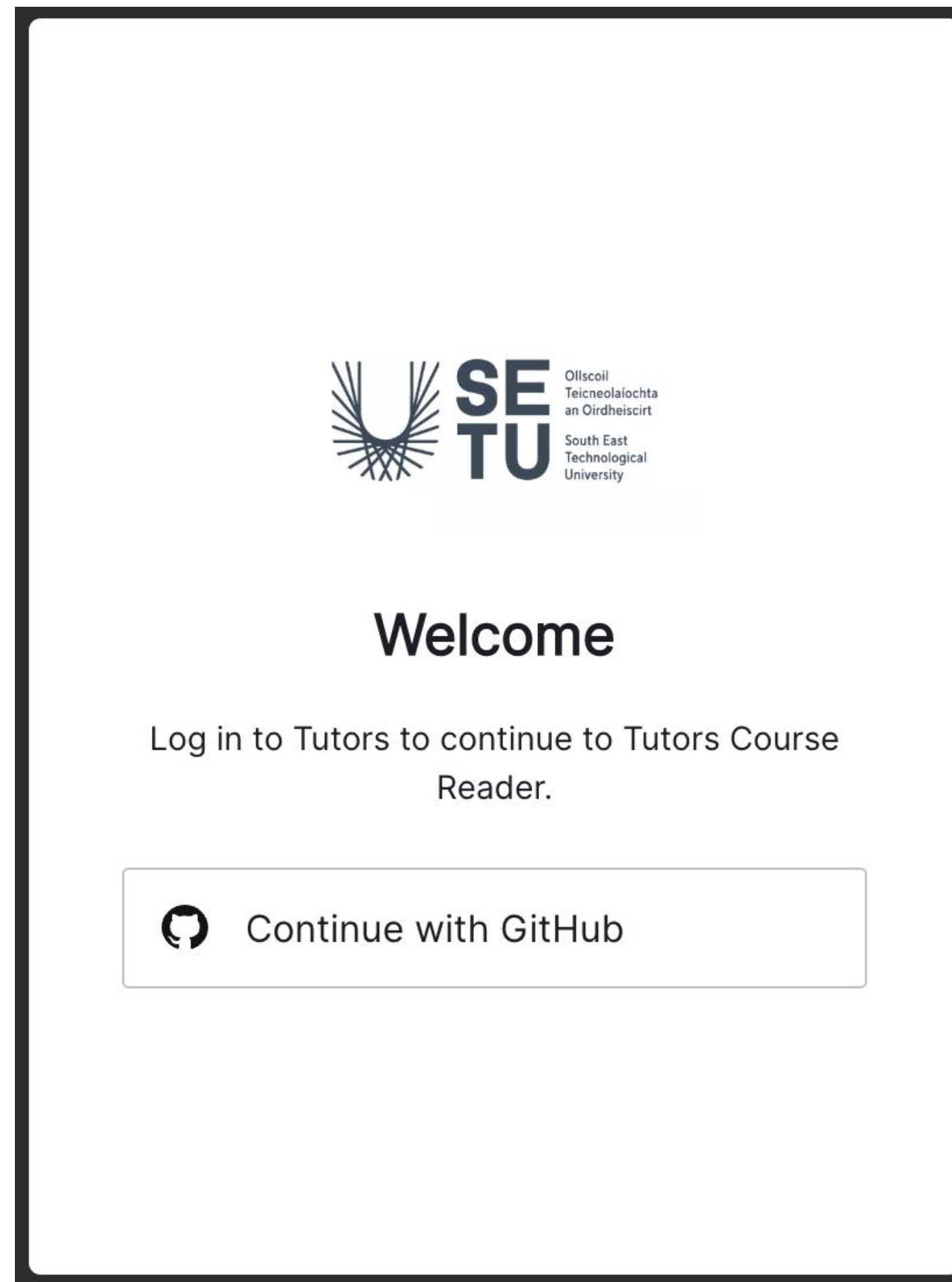
TutorsTime



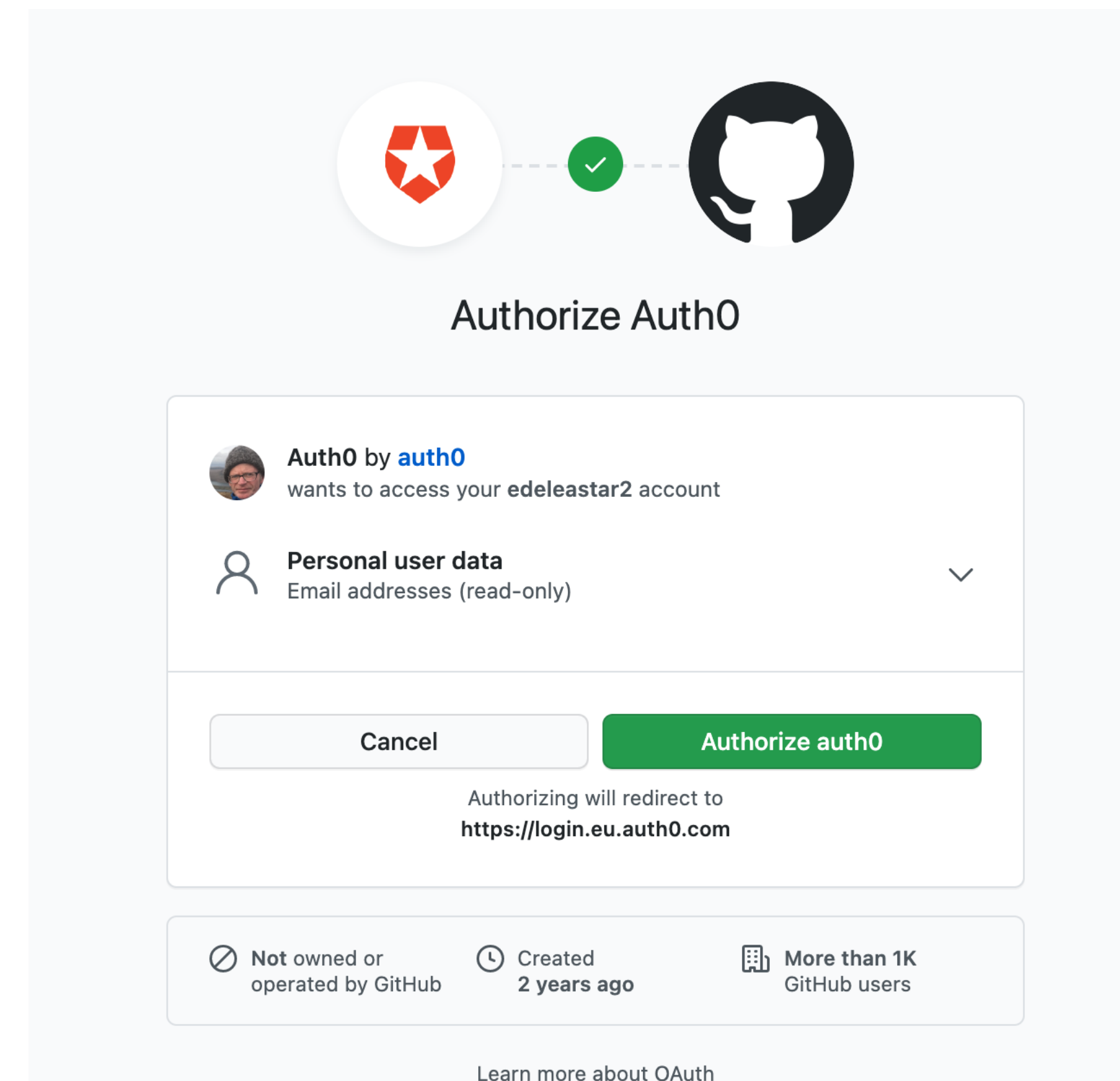
TutorsLive



When accessing module, you will be asked to sign in with Github



... first time you will have to authorise ...



TutorsTime

You have been authenticated via your GitHub credentials and are about to enter a Tutors course. Tutors will record how much time each view is active and send this to the TutorsTime data store. You can view this information via the TutorsTime feature on the course Navigator.

This data is available to you + the instructor of your course, but not to other students. No other data is gathered, nor is this data transmitted anywhere other than the TutorsTime data store.

TutorsLive also uses this data to display of panel of students currently online. You can opt out of this feature from the TutorsLive page.

To learn more about these features please consult:

- [Tutors Time FAQ](#)
- [Tutors Live FAQ](#)

Tutors is an open source application - the data collection component [is here](#).

You will be asked to authenticate again in seven days time.

[Proceed to Course](#)

Github Profile

 Search or jump to...  Pull requests Issues Marketplace Explore

 + 

Your profile picture has been updated. It may take a few moments to update across the site. 

 **edeleastar2**
Personal settings

Profile

Account

Appearance New

Account security

Billing & plans

Security log

Security & analysis

Public profile

Name

Eamonn de Leastar

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Public email

Select a verified email to display 

You have set your email address to private. To toggle email privacy, go to [email settings](#) and uncheck "Keep my email address private."

Bio

Profile picture



 Edit



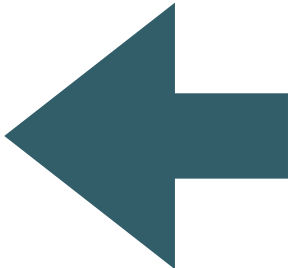
TutorsTime - Labs

Web Development
Eamonn de Leastar

Layout ▾

Labs Labs Chart Calendar

User	Github	Total	Date Last Accessed	Lab-00 Editing HTML	Lab-01 HTML Structure	Lab-02a CSS Fonts & Colours	Lab-02b CSS Paths & Links	Lab-03a Layout	Lab-03b Multicolumn	Lab-04 Navigation	Lab-05a Harp & Surge	Lab-05b Templates	Lab-06 Nav Params	Lab-07a Semantic-UI	Lab-07b Play Setup	Lab-08 Playlist 1
Eamonn de Leastar	edeleastar	1337	2/01/2022, 10:12:3	50	8	11	2	16	28	22	22	6	41	2	128	47



Layout ▾ 

Logged in as:
Eamonn de Leastar

Share Presence

View 0 Online

Tutors Live

Tutors Time

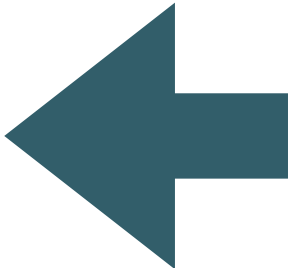
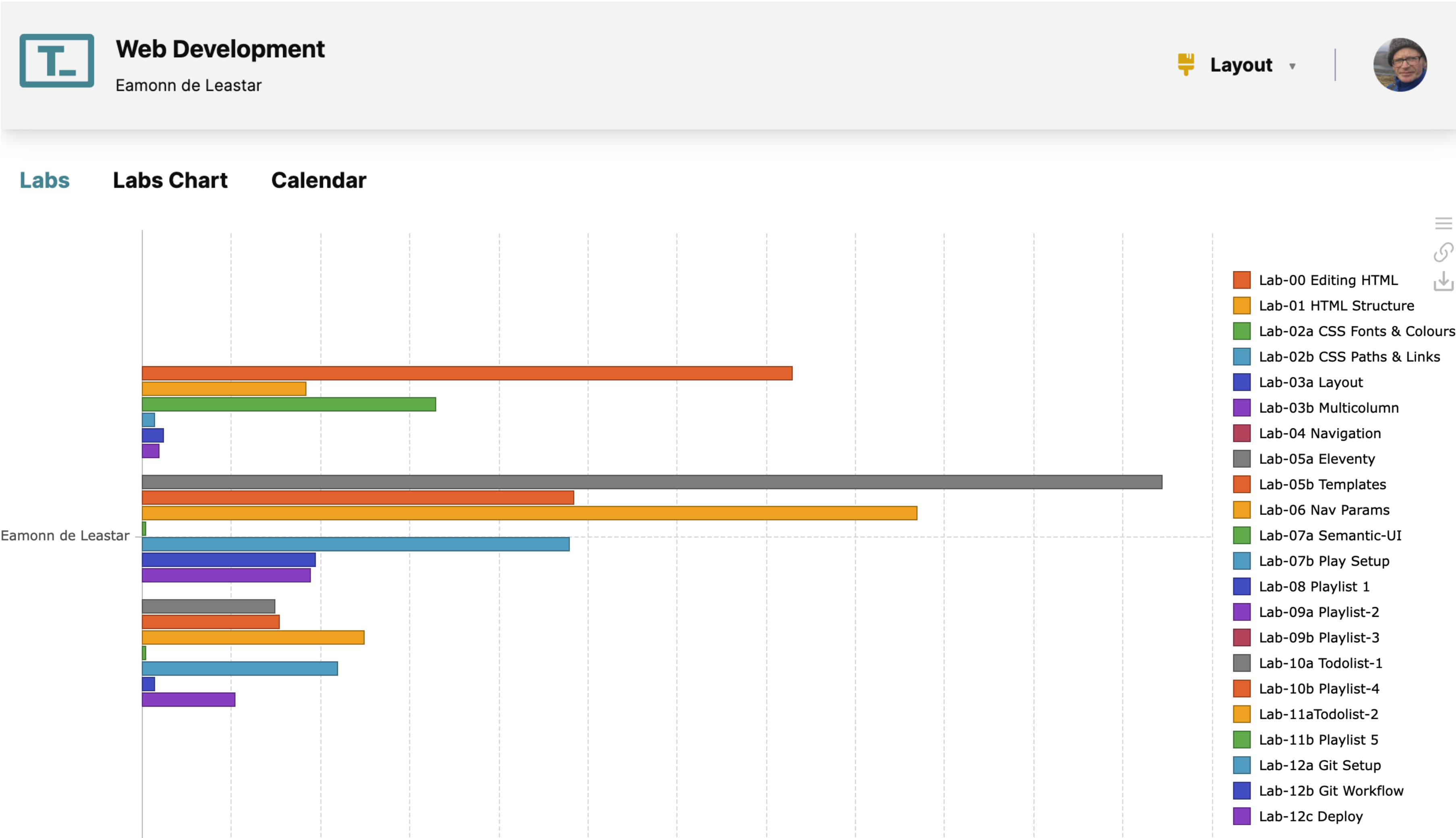
Github Profile

Logout

Number of minutes the Each lab has been active in your browser (estimate)



TutorsTime - Chart



Layout

Logged in as:
Eamonn de Leastar

Share Presence

View Online

Tutors Live

Tutors Time

Github Profile

Logout

At a glance
comparative time
spent on each
labs



TutorsTime - Calendar

T

Web Development

Eamonn de Leastar

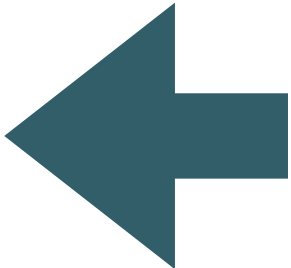
Layout

Labs

Labs Chart

Calendar

User	Github	Day	-1: Jan	0: Jan	1: Jan	2: Jan	3: Feb	Reading 1: Feb	4: Feb	5: Feb	6: Mar	Reading 2: Mar	7: Mar	8: Mar	9: Mar	Reading 3: Apr	
Eamonn de Leastar																	
		Mon	1	4		6	2	109		4	21	2		61	5	18	
		Tue	57		12	44	54			46	46					42	
		Wed			164	64	6			25	82	51		76	53		
		Thur	2		505	70	327	10	157	192	380	156	258	546	62	13	
		Fri	4	1	38	11	19		2	122	302	18	9	111	10	2	
		Sat	36		14	7	4				8				138	41	
		Sun			16	17			1			6			14	1	



Layout

0

Logged in as:
Eamonn de Leastar

Share Presence

View 0 Online

Tutors Live

Tutors Time

Github Profile

Logout

Time online each week of the semester

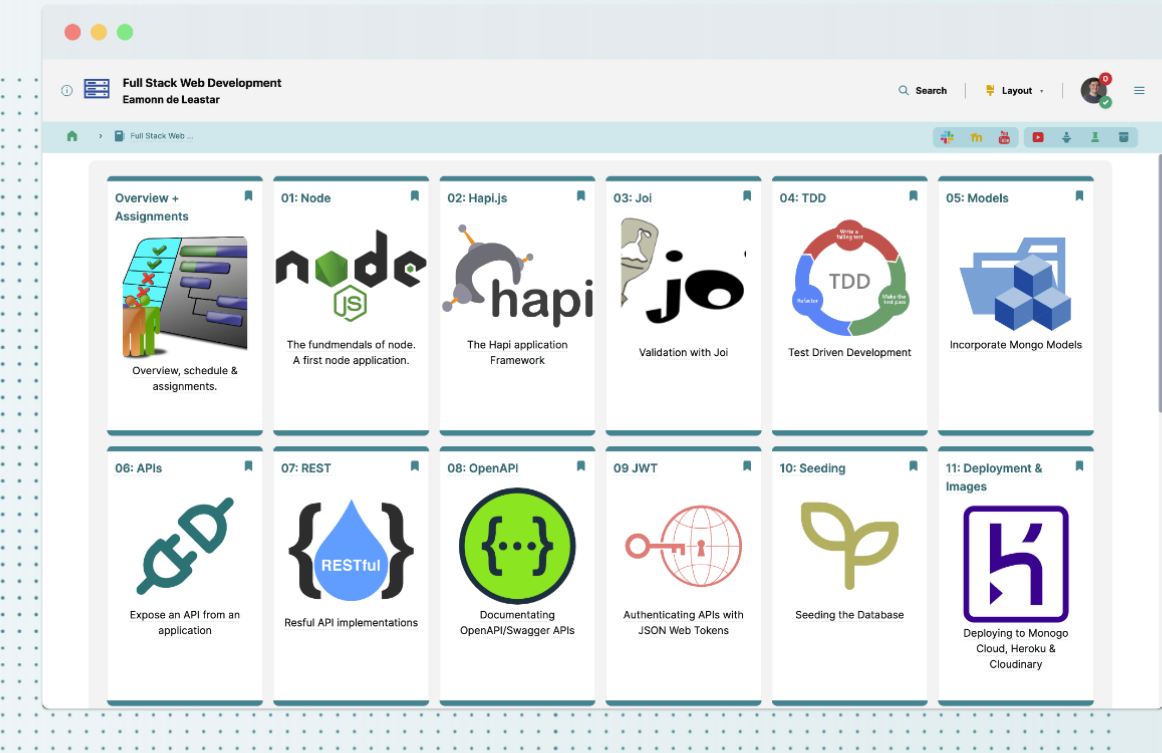


An Open Learning Web Toolkit

A collection of open source components & services supporting the creation of transformative learning experiences using open web standards.

View Demo

Documentation



<https://tutors.dev>

The Values of the project

Learner Experience



The **Learner Experience** prioritises web interactions that are **engaging, contextual, linkable, searchable, accessible and responsive**. In addition the experience should foster a sense of **community** and **connection** among fellow learners.

Educator Experience



The **Educator Experience** prioritises the creation of a **guided paths** through a curriculum via the creation of learning materials that are **autonomous, structurally aligned, composable, auditable, extensible, versioned** and **independent**.

Developer Experience



The **Developer Experience** prioritises the specification and implementation of **robust, well documented, loosely coupled components & services**, integrated into a **coherent toolkit** open to contributions from **diverse skill sets**.



<https://reader.tutors.dev/course/tutors-project-site>

Tutors Workshop



First contact with the
Tutors Tool Set

Learner Experience



Consuming a Tutors
Course

Educator Experience



Creating tutors courses

Developer Experience



The Tutors Open Source
Components & Services